



SCAFFOLD DESIGN REPORT

NIGERIA LIQUEFIED NATURAL GAS LIMITED



LOCATION: TRAIN 3	AREA: D
PERMIT TO WORK No.: —	TYPE OF SCAFFOLD: A Frame
SCAFFOLD DRAWING No: PMS-WA-ARCO-031	
RISK CLASSIFICATION: ☐ High ☒ Medium ☐ Low	

REFERENCE STANDARDS

- BS EN 1991-1-4:2005 — Wind Actions
- BS EN 1993-1-1:2005 — Design of Steel Structures (Eurocode 3)
- BS EN 12811-1:2003 — Temporary Works Equipment — Scaffolds

SCAFFOLD TO BE INSPECTED & TAGGED BEFORE USE

BRIEF DESCRIPTION OF SCAFFOLD DESIGN

This document describes the structural design of an **A Frame** scaffold required for *valve lifting during the turnaround maintenance activity in train 3* with equipment tag **311TRCA-015**. The scaffold system consists of a **1.8m × 1.8m × 3.5m** A-Frame scaffold with vertical bracing for sway resistance.

SAFE WORKING LOAD (SWL): 3.139 kN (320 kg)

TOTAL HORIZONTAL LOAD SUMMARY

Direction	Wind (kN)	Impact (kN)	TOTAL (kN)
X	1.88	0.942	2.822
Z	1.28	0.942	2.222

Role	ID	Name	Signature	Date
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	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	A Frame	Calculation Sheet	
	Document Number:	PMS-WA-ARCO-031	Prepared By: Nnamani, Ifeanyi	

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	Project Name:	SCAFFOLD DESIGN REPORT		
	Scaffold Type:	A Frame	Calculation Sheet	
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GENERAL NOTE

BS EN 12811-1, 4.2.1.2 Scaffold materials are in accordance with BS EN 12811 with minimum design strength of 210 N/mm². This design calculation was carried out in accordance with the limit states design method stipulated in EN 1993-1-1:2005 (Eurocode 3 – Design of Steel Structures).

Material Properties

48.3 mm outside diameter × 3.6 mm thick steel tube was defined by STAAD.Pro Connect Edition V22 section tables (British) and assigned to all scaffold members as section **PIP483.6**.

Modelling Philosophy

BS EN 12811-1, 4.2.1.2 A 3-D model was analysed using **STAAD.Pro CONNECT Edition (V22)** in compliance with EN 1993-1-1:2005. Base supports are modelled as pinned with lateral spring stiffness representing frictional restraint ($K_{FX}/K_{FZ} = 9.87 \text{ kN}$). Base supports are modelled as elastic spring supports, restrained laterally in both horizontal directions with rotational degrees of freedom released, representing flexible ground contact conditions at each standard foot. See [Appendix A](#) for model views.

LOAD CASES

Load diagrams are shown in [Appendix B](#).

Dead Load / Self-weight Self-weight of structural elements computed by STAAD.Pro with steel density **76.8195 kN/m³**, with a **10% allowance** for fittings and couplers (selfweight factor = 1.1).

Chain Hoist Load

BS EN 1991, 6.3.4.1 The equipment static weight is **256 kg ≈ 2.511 kN**. A **25% dynamic allowance** was applied for chain hoist operation.

Static weight = $256 \text{ kg} \times 9.81 / 1000 = \mathbf{2.511 \text{ kN}}$
Chain Hoist Design Load (CL) = $2.511 \times (1 + 0.25) = \mathbf{3.139 \text{ kN}}$

Impact Load

BS EN 1991, 6.3.4.1 30% of the chain hoist design load was assumed as a horizontal impact load, applied at the hoist attachment point independently in X and Z directions.

Impact Load (ILX) = $3.139 \times 0.30 = \mathbf{0.942 \text{ kN}}$ (X-axis)
Impact Load (ILZ) = $3.139 \times 0.30 = \mathbf{0.942 \text{ kN}}$ (Z-axis)

Wind Load

EN 1991-1-4:2005 Wind loads were derived in accordance with EN 1991-1-4:2005 using NLNG Bonny Island site parameters. Full calculation chain is given in the Wind Load Calculation sheet below. Wind UDL = **0.055202 kN/m** applied as a member UDL to all exposed scaffold members in the global X and Z wind directions.

Total Wind Load — X Axis (WLX) = **1.88 kN**
Total Wind Load — Z Axis (WLZ) = **1.28 kN**

WIND LOAD CALCULATION — EN 1991-1-4:2005

Reference	Calculation	Value	Unit
	Basic wind speed (3-second gust)		
	$V_3 =$	36.0	m/s
	<i>Eurocode $V_{b,0}$ is the characteristic 10-minute mean wind velocity</i>		
Dredger P. — How to Approximate	Convert 3-second gust to 10-minute mean wind velocity		
	$V_3 / V_{3600} =$	1.52	
	$V_{600} / V_{3600} =$	1.06	
	$V_{600} = (V_{600}/V_{3600}) \times (V_{3600}/V_3) \times V_3 =$	25.11	m/s
	Basic wind speed $V_b = V_{600} =$	25.11	m/s
EN 1991-1-4:2005	Roughness Length (z_0) — Terrain Category 0	0.003	m
	Terrain category II ($z_{0,ii}$)	0.05	
EN 1991-1-4, Table 4.1	z_{min}	1.0	m
	z — height of structure under consideration Source: STAAD command geometry height	3.5	m
EN 1991-1-4, Sec 4.3	$C_{r0}(z)$ — Roughness factor	1.0	
EN 1991-1-4, Sec 4.3	$C_o(z)$ — Orography factor	1.0	
EN 1991-1-4, Sec 4.3	Terrain factor $K_r = 0.19 \times [z_0/z_{0,ii}]^{0.07}$	0.156	
EN 1991-1-4, Table 4.1	$z_{min} < z \rightarrow$ Case 1 applies		
	Wind Turbulence		
	$C_r(z) = K_r \times \ln(z/z_0)$	1.1019	
	Mean velocity $V_m(z) = C_r(z) \times C_o(z) \times V_b$	27.664	m/s
EN 1991-1-4, Sec 4.4	K_i	1.0	
	Turbulence intensity $I_v(z) = k_i / [C_o(z) \times \ln(z/z_0)]$	0.1416	
	Peak Velocity Pressure		
	Density of air ρ	1.25	kg/m ³
EN 1991-1-4, Sec 4.5	$q_p(z) = [1 + 7 \cdot I_v(z)] \times 0.5 \times \rho \times V_m(z)^2$	952.4103	N/m ²
Wind Load = $C_f \times q_p \times A_f$			
$C_f =$		1.2	
$q_p =$		0.95241	kN/m ²
$A_f =$	Tube outer diameter (48.3 mm)	0.0483	m
Wind Load UDL =		0.055202	kN/m

LOAD COMBINATIONS

BS EN 12811-1, 6.2.9.2

Load combinations in accordance with BS EN 12811-1:2003. ULS combinations use $\gamma = 1.5$; SLS (characteristic) combinations use $\gamma = 1.0$.

DL = Dead Load | **CL** = Chain Hoist | **ILX/ILZ** = Impact X/Z | **WLX/WLZ** = Wind X/Z

LC No.	Combination	Component Loadings & Factors
7	1.5DL + 1.5CL	LC1 $\times$ 1.5 + LC2 $\times$ 1.5
8	1.5DL + 1.5CL + 1.5ILX	LC1 $\times$ 1.5 + LC2 $\times$ 1.5 + LC3 $\times$ 1.5
9	1.5DL + 1.5CL + 1.5ILZ	LC1 $\times$ 1.5 + LC2 $\times$ 1.5 + LC4 $\times$ 1.5
10	1.5DL + 1.5WLX	LC1 $\times$ 1.5 + LC5 $\times$ 1.5
11	1.5DL + 1.5WLZ	LC1 $\times$ 1.5 + LC6 $\times$ 1.5

LC No.	Combination	Component Loadings & Factors
12	1.5DL + 1.5CL + 1.5WLX	$LC1 \times 1.5 + LC2 \times 1.5 + LC5 \times 1.5$
13	1.5DL + 1.5CL + 1.5WLZ	$LC1 \times 1.5 + LC2 \times 1.5 + LC6 \times 1.5$
14	DL + CL	$LC1 \times 1.0 + LC2 \times 1.0$
15	DL + CL + ILX	$LC1 \times 1.0 + LC2 \times 1.0 + LC3 \times 1.0$
16	DL + CL + ILZ	$LC1 \times 1.0 + LC2 \times 1.0 + LC4 \times 1.0$
17	DL + WLX	$LC1 \times 1.0 + LC5 \times 1.0$
18	DL + WLZ	$LC1 \times 1.0 + LC6 \times 1.0$
19	DL + CL + WLX	$LC1 \times 1.0 + LC2 \times 1.0 + LC5 \times 1.0$
20	DL + CL + WLZ	$LC1 \times 1.0 + LC2 \times 1.0 + LC6 \times 1.0$

STATIC CHECK PARAMETERS — UTILIZATION RATIO SUMMARY

Maximum Utilization Ratio = 0.333 at Member No. **121**, Load Combination **10**, Clause **EC-6.3.3-662**. **PASS** — All members adequate; UC ratio below unity.

UC ratio diagram: see [Appendix C](#).

Member	Section	Status	Clause	UC Ratio	L/C	F_x (kN)	Type
121	PIP483.6	PASS	EC-6.3.3-662	0.333	10	0.89	C
38	PIP483.6	PASS	EC-6.3.3-662	0.282	12	1.19	C
28	PIP483.6	PASS	EC-6.2.9.1	0.278	12	0.57	T
24	PIP483.6	PASS	EC-6.2.9.1	0.259	10	0.15	T
96	PIP483.6	PASS	EC-6.2.9.1	0.224	8	4.26	C
59	PIP483.6	PASS	EC-6.3.3-662	0.223	12	0.22	C
90	PIP483.6	PASS	EC-6.3.3-662	0.210	9	1.69	C
104	PIP483.6	PASS	EC-6.2.9.1	0.208	8	4.40	C
89	PIP483.6	PASS	EC-6.3.3-662	0.202	9	1.71	C
82	PIP483.6	PASS	EC-6.2.9.1	0.201	12	0.58	C
105	PIP483.6	PASS	EC-6.2.9.1	0.201	13	0.22	T
102	PIP483.6	PASS	EC-6.2.9.1	0.200	8	0.31	T
45	PIP483.6	PASS	EC-6.2.9.1	0.199	8	0.01	T
80	PIP483.6	PASS	EC-6.2.9.1	0.199	9	0.69	C
86	PIP483.6	PASS	EC-6.3.3-662	0.199	8	1.54	C
112	PIP483.6	PASS	EC-6.2.9.1	0.199	8	0.01	T
99	PIP483.6	PASS	EC-6.2.9.1	0.191	9	0.13	T
69	PIP483.6	PASS	EC-6.2.9.1	0.189	9	1.86	T
95	PIP483.6	PASS	EC-6.2.9.1	0.185	8	0.93	C
65	PIP483.6	PASS	EC-6.2.9.1	0.183	8	0.36	T
75	PIP483.6	PASS	EC-6.2.9.1	0.180	8	4.23	T
15	PIP483.6	PASS	EC-6.3.3-662	0.178	12	6.31	C
68	PIP483.6	PASS	EC-6.2.9.1	0.178	9	2.00	T
111	PIP483.6	PASS	EC-6.2.9.1	0.176	12	0.04	T
108	PIP483.6	PASS	EC-6.2.9.1	0.175	13	0.20	T
103	PIP483.6	PASS	EC-6.2.9.1	0.171	8	4.37	T
40	PIP483.6	PASS	EC-6.3.3-662	0.168	10	0.34	C
78	PIP483.6	PASS	EC-6.2.9.1	0.166	12	0.56	C
85	PIP483.6	PASS	EC-6.3.3-662	0.162	9	1.27	C
13	PIP483.6	PASS	EC-6.2.9.1	0.151	10	3.13	T
74	PIP483.6	PASS	EC-6.2.9.1	0.151	8	0.90	T

Member	Section	Status	Clause	UC Ratio	L/C	F _x (kN)	Type
64	PIP483.6	PASS	EC-6.2.9.1	0.148	8	0.85	T
57	PIP483.6	PASS	EC-6.3.3-662	0.142	10	3.24	C
49	PIP483.6	PASS	EC-6.2.9.1	0.136	12	0.04	T
61	PIP483.6	PASS	EC-6.3.3-662	0.135	8	0.96	C
76	PIP483.6	PASS	EC-6.2.9.1	0.135	13	0.56	C
17	PIP483.6	PASS	EC-6.2.9.1	0.131	12	1.35	C
101	PIP483.6	PASS	EC-6.2.9.1	0.126	8	2.30	C
110	PIP483.6	PASS	EC-6.2.9.1	0.125	8	2.62	C
19	PIP483.6	PASS	EC-6.2.9.1	0.121	10	1.54	C
36	PIP483.6	PASS	EC-6.3.3-662	0.117	12	4.77	C
97	PIP483.6	PASS	EC-6.2.9.1	0.117	9	0.67	T
98	PIP483.6	PASS	EC-6.2.9.1	0.117	9	1.40	C
100	PIP483.6	PASS	EC-6.2.9.1	0.116	9	2.58	T
106	PIP483.6	PASS	EC-6.2.9.1	0.110	13	2.56	T
55	PIP483.6	PASS	EC-6.2.9.1	0.106	10	0.60	T
81	PIP483.6	PASS	EC-6.2.9.1	0.105	9	0.04	T
124	PIP483.6	PASS	EC-6.3.3-661	0.105	9	1.47	C
109	PIP483.6	PASS	EC-6.3.3-662	0.101	9	3.30	C
107	PIP483.6	PASS	EC-6.2.9.1	0.099	8	2.62	C
34	PIP483.6	PASS	EC-6.3.3-662	0.096	7	2.10	C
77	PIP483.6	PASS	EC-6.2.9.1	0.095	8	0.02	C
125	PIP483.6	PASS	EC-6.3.3-661	0.089	12	1.15	C
70	PIP483.6	PASS	EC-6.2.9.1	0.078	10	0.26	T
66	PIP483.6	PASS	EC-6.2.9.1	0.076	10	0.33	T
120	PIP483.6	PASS	EC-6.2.9.1	0.073	12	0.35	T
87	PIP483.6	PASS	EC-6.2.9.1	0.069	10	0.01	T
91	PIP483.6	PASS	EC-6.2.9.1	0.068	10	0.08	C
84	PIP483.6	PASS	EC-6.3.3-662	0.046	8	2.38	C
113	PIP483.6	PASS	EC-6.3.3-662	0.035	8	1.38	C
114	PIP483.6	PASS	EC-6.3.3-662	0.032	12	1.22	C

SCAFFOLD TUBES CONNECTION STABILITY CHECK

EN 12811, Table C.1

The maximum axial load on scaffold tube members (from STAAD.Pro Connect Edition analysis) is **3.526 kN** (Member 96, C). **Right-angle double coupler of Class A (10 kN slipping resistance)** shall be used for all scaffold steel tube connections at node joints.

Check: 3.526 kN < 10.0 kN → **PASS**

Member force table: see [Appendix D](#).

Table C.1 — Characteristic Values of Resistances for Couplers (EN 12811)

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Right-angle Coupler (RA)	Slipping force F _{s,k} (kN)	10.0	15.0
	Cruciform bending moment M _{B,k} (kNm)	—	0.8
	Pull-apart force F _{p,k} (kN)	20.0	30.0
	Rotational moment M _{T,k} (kNm)	—	0.13

Coupler Type	Resistance	Characteristic Value	
		Class A	Class B
Swivel Coupler (SW)	Slipping force $F_{s,k}$ (kN)	6.0	9.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.8
	Pull-apart force $F_{p,k}$ (kN)	15.0	22.5
	Rotational moment $M_{T,k}$ (kNm)	—	0.08
Putlog / Single Coupler (PL)	Slipping force $F_{s,k}$ (kN)	—	3.0
	Cruciform bending moment $M_{B,k}$ (kNm)	—	0.6

FRICTIONAL RESISTANCE

Max vertical reaction per support (Max f_y) = Resistance $\times$ 0.15 / C_F	= 4.935 kN
Coefficient of friction (C_F)	= 0.3
Frictional Resistance (FR) = $C_F \times$ Max f_y	= 1.4805 kN
Spring Stiffness = FR / 0.15	= 9.87 kN ← applied as KFX/KFZ in STAAD model

DEFLECTION CHECK RESULTS

BS EN 12811-1, Cl 6.3 Serviceability limit state — values from STAAD.Pro SLS (unfactored) combination results. Diagrams and displacement tables are in [Appendix C](#) and [Appendix D](#).

Vertical Deflection Check

Parameter	Value	Allowable	Check
Max vertical displacement — LC 17 Member 28 Member length L = 1800.0 mm	1.875 mm	$L/100 = 1800/100 = 18.0$ mm	PASS

Horizontal Deflection Check

Parameter	Value	Allowable	Check
Max horizontal displacement — LC 15 Member 38 Member length L = 1450.0 mm	1.349 mm	$L/200 = 1450/200 = 7.2$ mm	PASS

CONCLUSION

The structural analysis and design of the **A Frame** scaffold confirms the structure is adequate for its intended purpose. All member utilization ratios are below unity (maximum UC = **0.333** at Member 121). Vertical and horizontal deflections are within their respective allowable limits of **L/100 = 18.0mm** and **L/200 = 7.2mm**. All scaffold tube coupler connections are adequate for the applied axial loads.

RECOMMENDATIONS

The scaffold shall be inspected by a competent Scaffold Inspector after erection to confirm conformity with the design specification and approved working drawings before use. All erection work shall comply with the design intent and material specification. Any proposed change to the design requires a re-design and the Scaffold Engineer shall be duly informed.

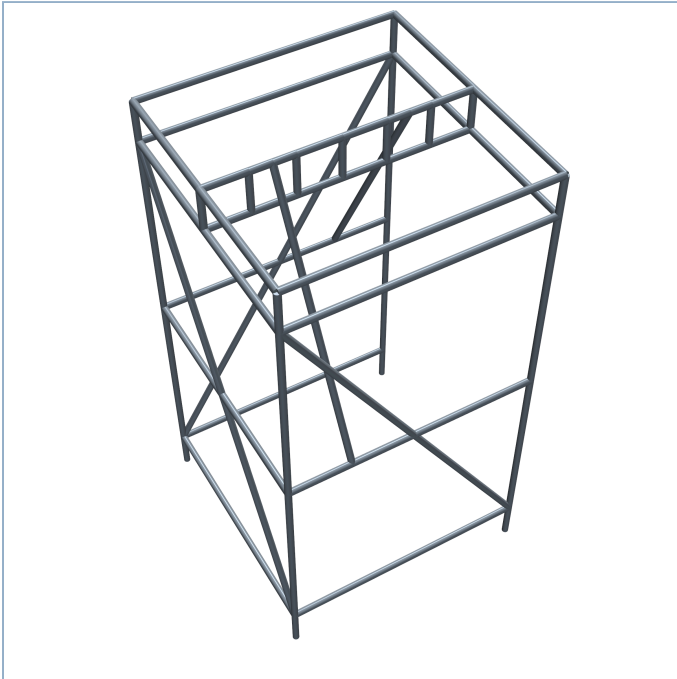
INSTALLATION NOTES

- The scaffold working drawings and purpose of the scaffold must be reviewed and discussed by all personnel involved prior to execution.

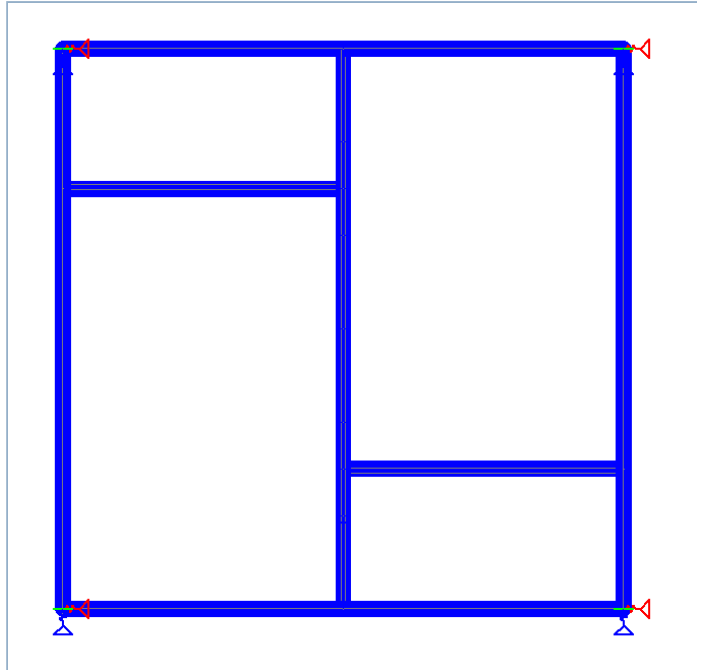
2. The work vicinity must be barricaded and appropriate signage placed at all access points.
3. All scaffolders must wear complete PPE in accordance with NLNG work-at-height regulations.
4. A valid work permit must be obtained, discussed, and signed before erection commences.
5. Scaffold erection must be carried out in strict accordance with NASC guidance and site safety procedures.
6. Lay out and level the four base plates at the standard positions, ensuring a plan footprint of 1.8m x 1.8m in accordance with the approved working drawing.
7. Erect the four scaffold tube standards (48.3 x 3.6 mm) vertically on the base plates and secure with base collars. Check plumb in both axes before proceeding.
8. Install the lower horizontal ledgers and transoms at the kicker lift height (0.300m) to tie the four standards together and form the rigid base frame.
9. Install mid-level horizontal ledgers and transoms at 1.8m height in accordance with the working drawing.
10. Install mid-level horizontal ledgers and transoms at 3.2m height in accordance with the working drawing.
11. Fix all vertical bracing members diagonally across the face panels of the A-frame structure using right-angle couplers to provide sway resistance in both X and Z axes as shown on the working drawing.
12. Install the 1800mm ladder beam horizontally across the top of the standards at 3.5m height, connecting to both standards with right-angle double couplers. Ensure the beam is level before fully tightening all couplers.
13. Rig the chain hoist to the ladder beam at the designated hoist point shown on the working drawing. Confirm the hoist is rated for a minimum safe working load equal to or greater than the design SWL.
14. The chain hoist attachment point and all coupler connections at ladder beam level shall be inspected and torque-checked by a competent person before any lifting operation commences.
15. The scaffold structure must be inspected and tagged by a competent advance scaffold inspector after erection and before any lift is performed.
16. The safe working load (SWL) stated on the design report must not be exceeded at any time during operations.
17. All modifications made after erection must be carried out by a competent scaffold team with the written approval of the scaffold engineer. End-user or third-party modification of this scaffold structure is strictly prohibited.
18. For dismantling, perform the erection steps in reverse order.

APPENDIX A — MODEL VIEWS

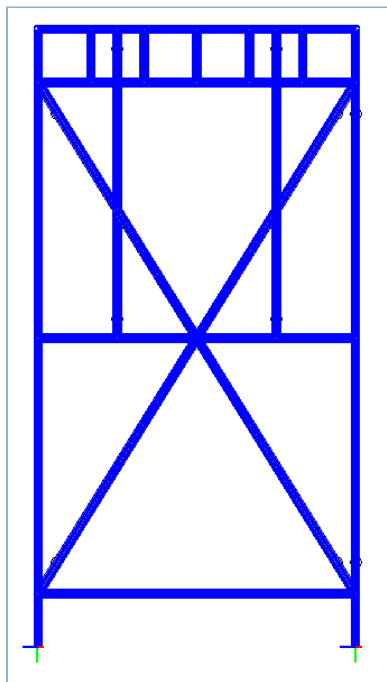
3D Isometric View



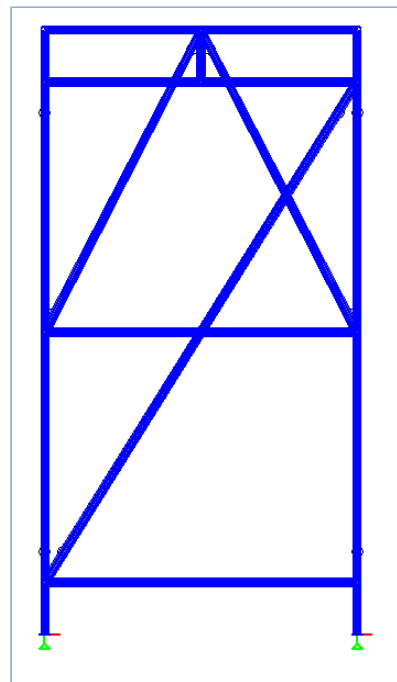
Plan View



Elevation from +X Axis



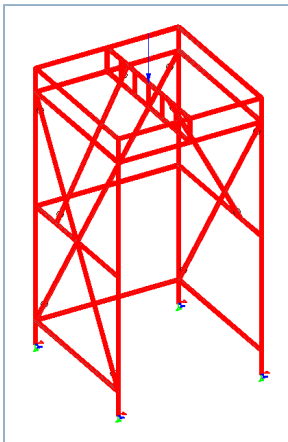
Elevation from +Z Axis



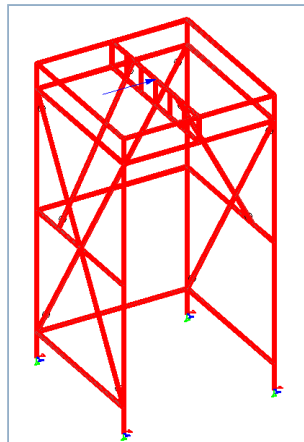
Scaffold Dimensions: W = 1.8m | D = 1.8m | H = 3.5m | **Lifts:** 0.3m, 1.75m, 3.2m, 3.5m

APPENDIX B — LOAD DIAGRAMS

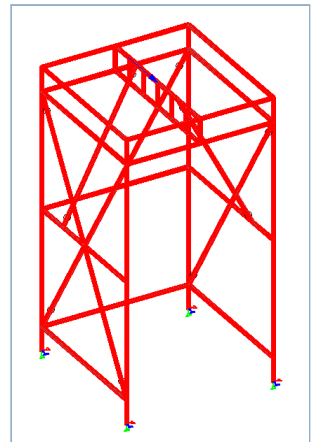
Chain Hoist Load (CL)



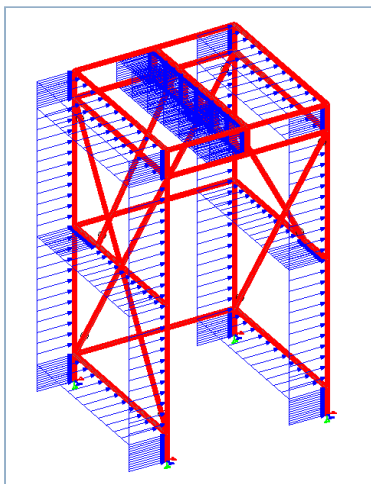
Impact Load — X Axis (ILX)



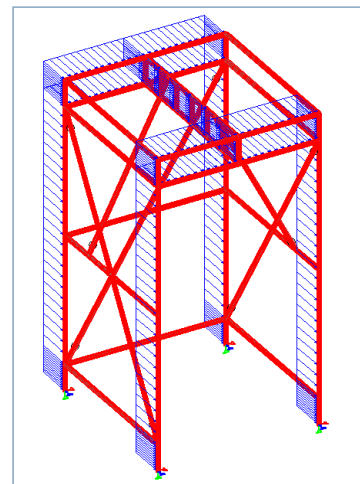
Impact Load — Z Axis (ILZ)



Wind Load — X Axis (WLX)

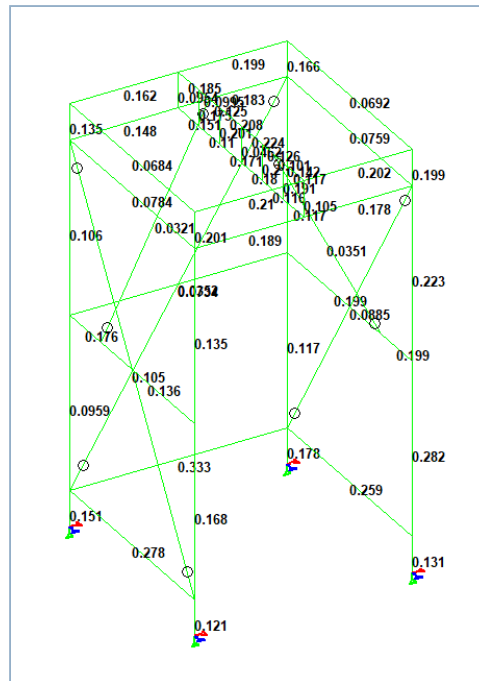


Wind Load — Z Axis (WLZ)

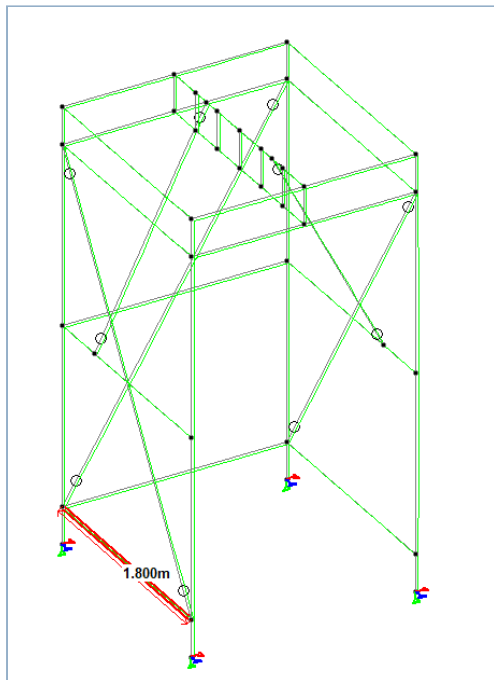


APPENDIX C — ANALYSIS RESULTS

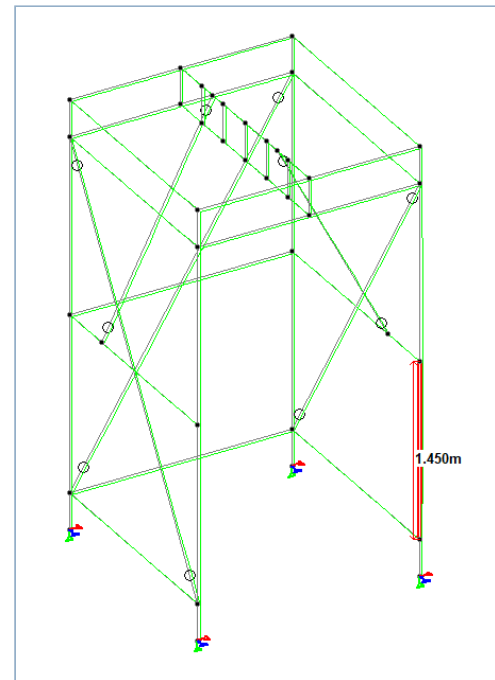
Utilization Ratio (UC) Diagram



Vertical Deflection Diagram



Horizontal Deflection Diagram



APPENDIX D — GUI RESULTS TABLES

Member Force / Code Check Table (Connection Stability)

Beam	L/C	Node	Axial Force kN	Shear-Y kN	Shear-Z kN	Torsion kN-m	Moment-Y kN-m	Moment-Z kN-m
96	16	45	3.526	-0.804	-0.082	-0.006	0.007	-0.165
96	20	45	3.104	-0.751	-0.083	-0.006	0.008	-0.157
96	14	45	3.092	-0.761	-0.083	-0.006	0.007	-0.158
75	20	48	3.080	0.763	0.030	0.006	0.008	-0.072
103	16	36	3.078	-0.778	0.028	0.006	-0.007	0.157
75	14	48	3.076	0.771	0.030	0.006	0.008	-0.074
104	14	51	3.041	0.825	-0.086	-0.006	0.027	0.080
75	16	48	3.040	0.814	0.030	0.006	0.008	-0.080
104	20	51	3.029	0.834	-0.085	-0.006	0.027	0.081
103	14	36	3.026	-0.821	0.028	0.006	-0.007	0.163
103	20	36	3.022	-0.829	0.027	0.006	-0.008	0.165
104	15	51	2.934	0.747	0.332	0.006	0.043	0.068
103	15	36	2.915	-0.736	-0.110	-0.006	-0.080	0.152
96	15	45	2.842	-0.828	-0.431	-0.021	0.150	-0.167
75	15	48	2.823	0.866	0.070	0.021	-0.052	-0.092
96	19	45	2.741	-0.774	-0.042	-0.009	0.017	-0.159
104	19	51	2.735	0.792	-0.017	-0.009	0.015	0.076
75	19	48	2.718	0.809	-0.024	0.009	-0.017	-0.083
103	19	36	2.713	-0.802	-0.049	0.009	-0.017	0.159
104	16	51	2.622	0.782	-0.085	-0.006	0.026	0.073
101	16	49	2.203	-0.900	-0.109	-0.002	-0.012	-0.123
109	16	54	2.203	-0.576	0.083	-0.002	-0.029	0.013
109	20	54	1.889	-0.526	0.080	-0.002	-0.028	0.012
101	20	49	1.889	-0.845	-0.109	-0.002	-0.012	-0.115
109	14	54	1.854	-0.533	0.083	-0.002	-0.029	0.013
101	14	49	1.854	-0.858	-0.110	-0.002	-0.012	-0.116
103	15	36	1.854	-0.858	-0.110	-0.002	-0.012	-0.116

Vertical Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
28	1.875	1.050	17	959	51.894	-0.046	4.642
28	1.835	1.050	19	981	51.228	0.040	4.505
24	1.812	0.750	17	993	52.672	-0.026	-4.635
24	1.709	0.750	19	1053	53.042	0.034	-4.505
28	1.108	1.200	15	1624	26.112	-0.005	2.789
24	0.943	0.750	15	1908	27.403	0.044	-2.776
121	0.901	0.450	19	1996	52.899	-0.428	3.044
121	0.779	0.450	17	2310	53.090	-0.475	2.936
120	0.733	0.750	19	2457	54.230	-0.105	1.464
121	0.665	0.450	15	2707	27.075	-0.220	2.024
120	0.650	0.900	15	2770	27.492	-0.105	0.661
45	0.645	0.900	19	2091	59.941	-0.944	-4.397
45	0.563	0.900	15	2399	31.689	-1.044	-2.696
45	0.525	0.787	17	2569	58.997	-0.501	-4.541
120	0.512	0.900	16	3513	-0.737	-0.076	24.849
120	0.497	0.900	20	3623	-0.758	-0.074	34.346
120	0.487	0.900	14	3692	-0.746	-0.082	0.474
49	0.412	0.787	19	3273	56.841	-0.836	4.401
49	0.408	0.675	17	3310	58.727	-0.259	4.562
49	0.405	0.900	20	3336	-1.714	-0.764	33.685
49	0.392	0.900	14	3440	-1.696	-0.763	-0.182
49	0.388	0.900	16	3474	-1.711	-0.746	24.170
121	0.307	1.050	16	5858	-0.201	-0.087	24.437
120	0.307	0.750	17	5861	54.931	-0.095	1.066
121	0.298	1.050	20	6036	-0.198	-0.090	33.221
121	0.295	1.050	14	6098	-0.202	-0.092	0.302

Max = 1.875 mm (LC 17) | L = Member 28 length (1800 mm) | Allowable L/100 = 18.0 mm → **PASS**

Horizontal Displacement Table (SLS Combinations)

Beam	Max Disp mm	Location m	L/C	L/Displ	Global X mm	Global Y mm	Global Z mm
38	1.597	0.846	19	907	58.900	-0.010	-4.164
38	1.367	0.846	17	1060	57.422	-0.000	-4.435
61	1.116	0.725	17	1298	63.512	-0.008	4.577
61	1.103	0.725	19	1314	62.366	-0.024	4.341
40	1.088	0.846	17	1332	55.161	-0.005	4.640
38	1.005	0.846	15	1442	31.465	-0.013	-2.413
59	0.991	0.604	17	1462	64.133	0.003	-4.720
59	0.975	0.604	19	1487	64.481	-0.015	-4.685
40	0.966	0.846	19	1500	53.625	-0.013	4.587
61	0.721	0.846	15	2010	32.706	-0.022	2.706
59	0.623	0.483	15	2325	34.549	-0.022	-3.001
36	0.503	0.725	17	2885	54.504	-0.028	-4.660
34	0.500	0.725	19	2898	54.002	-0.002	4.208
34	0.489	0.725	17	2967	54.496	0.007	4.540
36	0.479	0.725	19	3025	54.019	-0.035	-4.595
38	0.462	0.846	20	3141	1.621	-0.024	34.017
40	0.387	0.846	20	3746	-1.719	-0.015	33.667
38	0.354	0.846	16	4099	1.702	-0.025	24.668
40	0.336	0.846	15	4314	26.097	-0.011	2.836
55	0.302	0.483	20	4805	-0.643	-0.012	33.690
38	0.275	0.846	14	5267	1.615	-0.015	0.307
38	0.269	0.846	18	5381	0.144	-0.014	33.746
40	0.261	0.846	18	5559	-0.184	-0.007	33.719
40	0.260	0.846	16	5566	-1.778	-0.016	24.305
55	0.248	0.483	19	5856	53.993	-0.009	4.677
36	0.247	0.725	15	5867	27.478	-0.029	-2.886

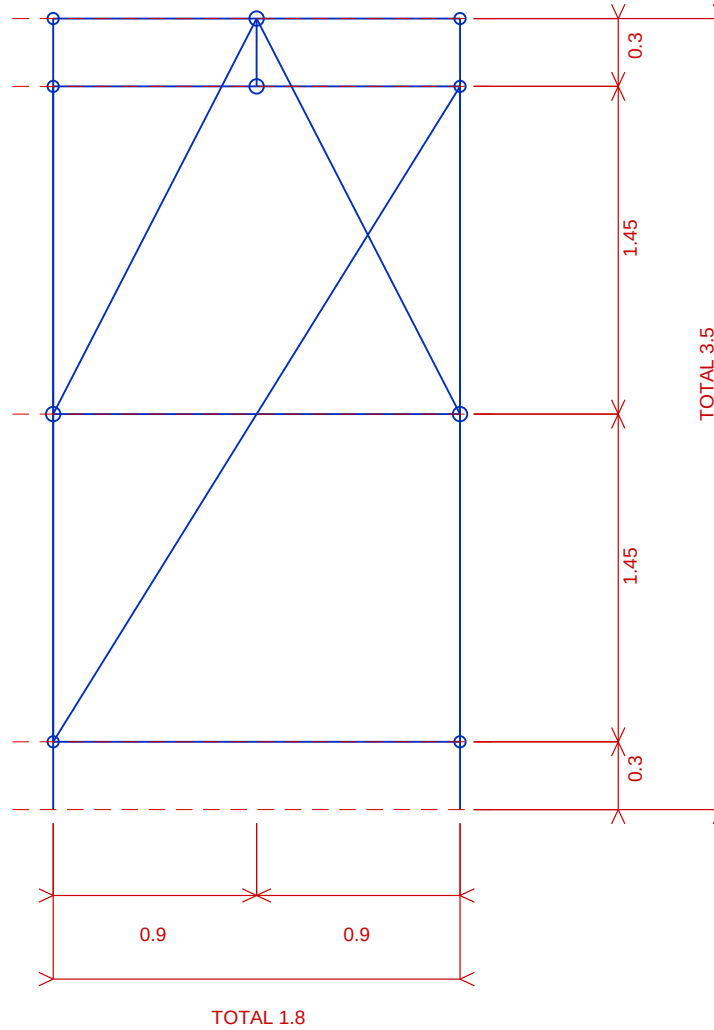
Max = 1.349 mm (LC 15) | L = Member 38 length (1450 mm) | Allowable L/200 = 7.2 mm → **PASS**

APPENDIX E — REFERENCES & STANDARDS

Reference	Title	Year
BS EN 12811-1:2003	Temporary Works Equipment — Scaffolds — Performance Requirements and General Design	2003
BS EN 1993-1-1:2005	Eurocode 3: Design of Steel Structures — General Rules and Rules for Buildings	2005
BS EN 1991-1-4:2005	Eurocode 1: Actions on Structures — Wind Actions	2005
BS EN 1990:2002	Eurocode: Basis of Structural Design	2002
NASC TG20:13	Guide to Good Practice for Scaffolding with Tubes and Fittings	2013
Dredger P.	How to Approximate 10-Minute Mean Wind Speed from 3-Second Gust	—
STAAD.Pro CONNECT	STAAD.Pro CONNECT Edition V22 — Structural Analysis and Design Software	2022

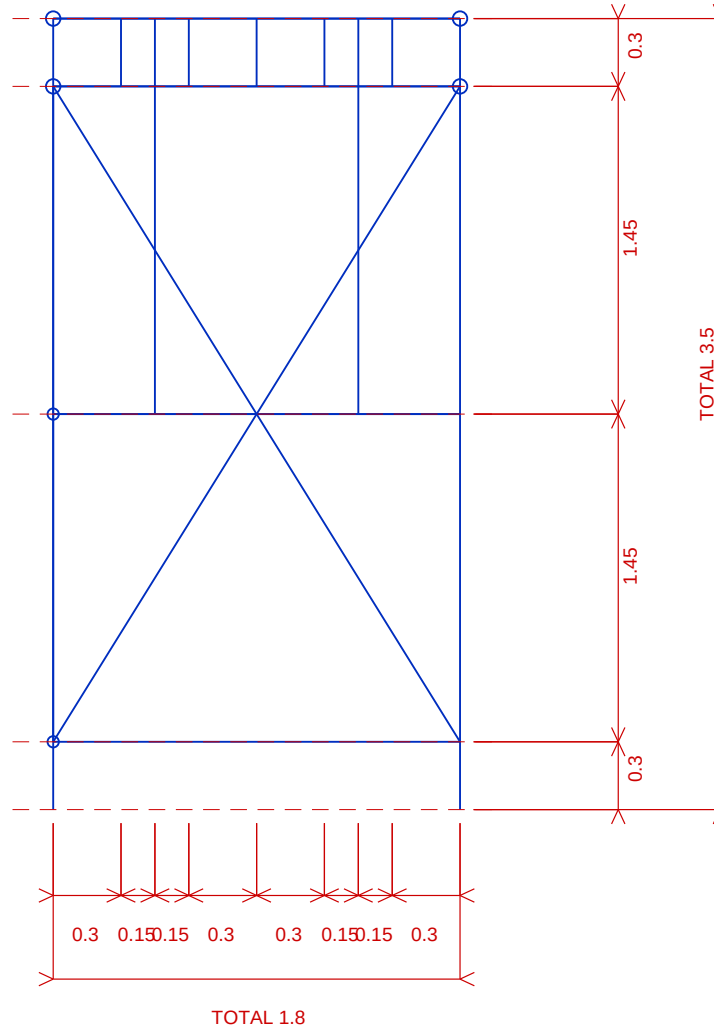
Software Used: STAAD.Pro CONNECT Edition V22 (Bentley Systems) — 3D Structural Analysis and Code Checking per EN 1993-1-1:2005.

Report: A-Frame Scaffold Structural Design Report — ARCO M&E, NLNG Bonny Island.



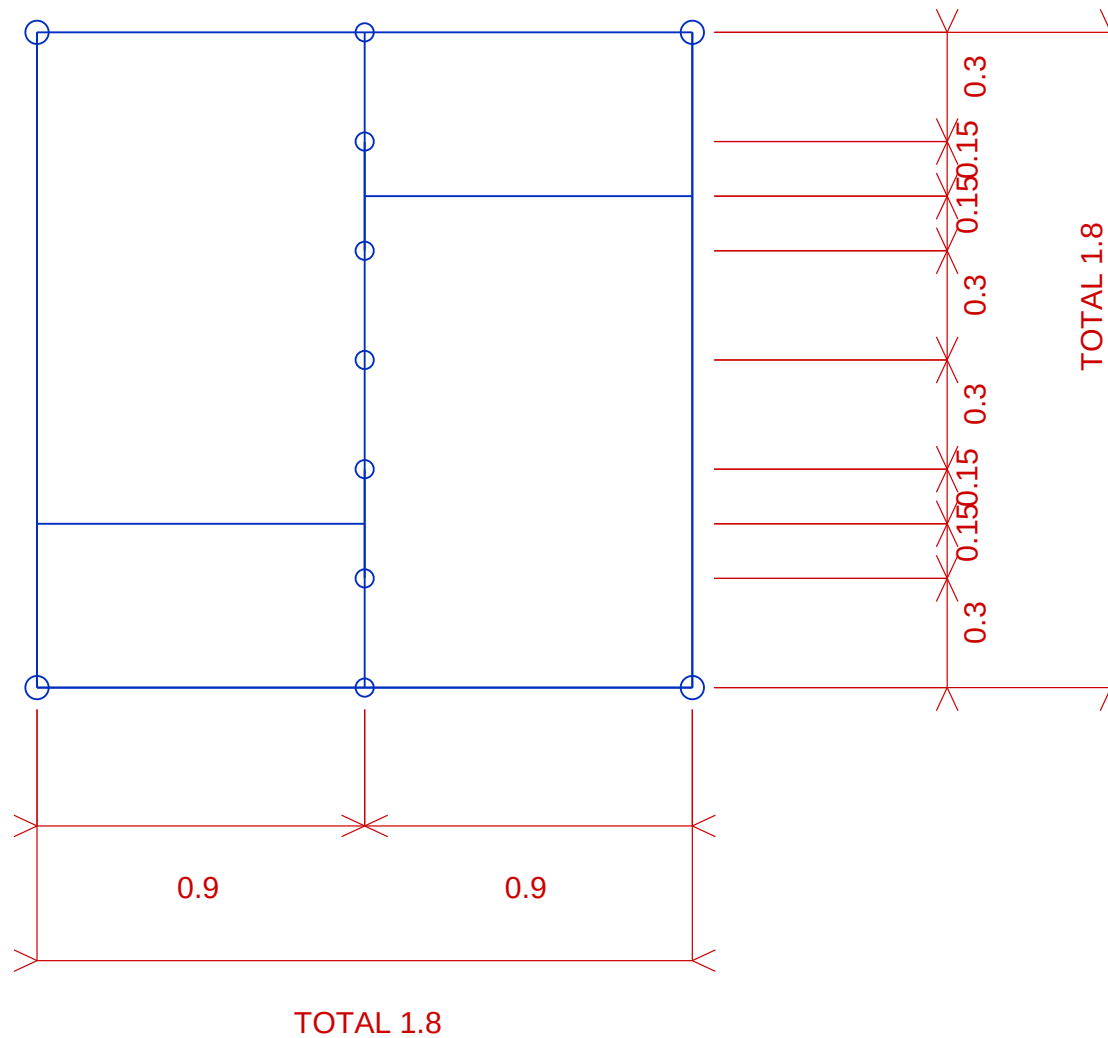
FRONT VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-FRONT-VIEW			 NLNG		
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - FRONT VIEW		SCALE NTS	DATE 08-JUN-2026	REV 0			
CLIENT		LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY	
						SHEET 1 OF 4	SIZE A4



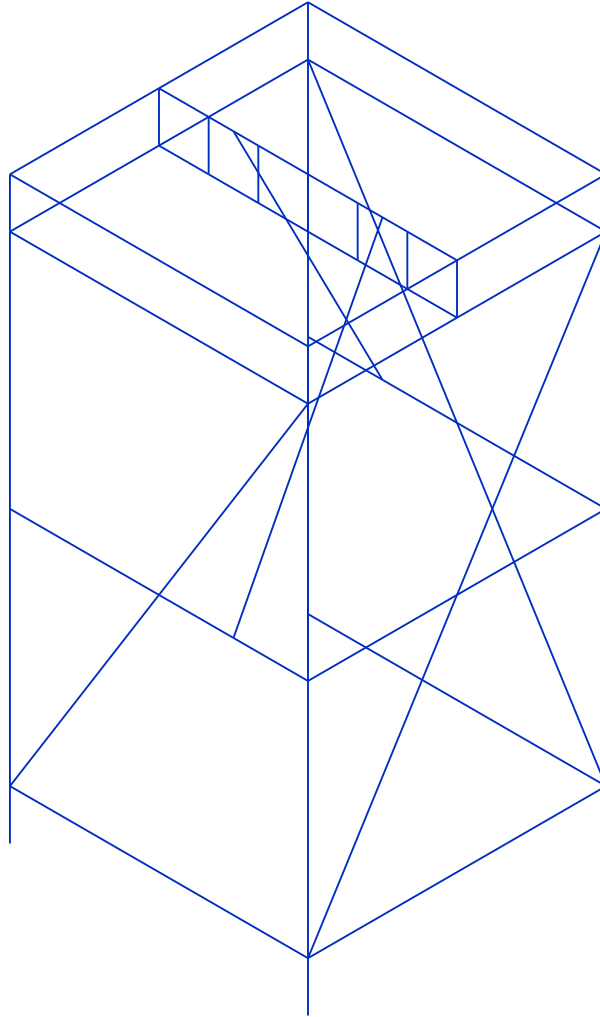
SIDE VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-SIDE-VIEW			 NLNG		
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - SIDE VIEW		SCALE NTS	DATE 08-JUN-2026	REV 0			
CLIENT		LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY	
						SHEET 2 OF 4	SIZE A4



PLAN VIEW

PROJECT		DRAWING NO.				
SCAFFOLDING WORKS		PMS-WA-ARCO-003-PLAN-VIEW				
DRAWING TITLE		SCALE	DATE	REV	COMPANY	
SCAFFOLD GENERAL ARRANGEMENT - PLAN VIEW		NTS	08-JUN-2026	0		
CLIENT	LOCATION	DRAWN	CHECKED	APPROVED	SHEET	
		Nnamani, Ifeanyi				
					3 OF 4	A4



3D VIEW

PROJECT SCAFFOLDING WORKS		DRAWING NO. PMS-WA-ARCO-003-3D-VIEW			 NLNG	
DRAWING TITLE SCAFFOLD GENERAL ARRANGEMENT - 3D VIEW		SCALE NTS	DATE 08-JUN-2026	REV 0		
CLIENT	LOCATION	DRAWN Nnamani, Ifeanyi	CHECKED	APPROVED	COMPANY	
					SHEET 4 OF 4	SIZE A4

3D Isometric View

